

SRM VEC
DEPARTMENT OF AIDS

Continuous Assessment Test - 1

(Regulations 2023)

AD3445 – AI1

Branch: B.Tech | Date: 01.03.2026 | Max. Marks: 100 | Duration: 3 Hrs

Answer ALL Questions

(100 Marks)

PART A — (10 × 2 = 20 marks)

1. Define Artificial Intelligence according to the 'Acting Rationally' approach. [CO1] [BTL1-Remember] (2)
2. Specify the PEAS description for an Automated Taxi Driver system. [CO1] [BTL2-Understand] (2)
3. Distinguish between a well-defined problem and a goal formulation. [CO2] [BTL2-Understand] (2)
4. What is the significance of the 'Admissibility' property in A* search? [CO2] [BTL2-Understand] (2)
5. Compare Breadth-First Search (BFS) and Depth-First Search (DFS) in terms of space complexity. [CO2] [BTL3-Apply] (2)
6. Define the term 'Heuristic Function' with an example. [CO2] [BTL1-Remember] (2)
7. What are the three components of a Constraint Satisfaction Problem (CSP)? [CO3] [BTL1-Remember] (2)
8. Explain the concept of 'Alpha-Beta Pruning' in game playing. [CO3] [BTL2-Understand] (2)
9. Identify the limitations of Hill Climbing search. [CO2] [BTL2-Understand] (2)
10. What is a 'Utility Function' in the context of an agent? [CO1] [BTL1-Remember] (2)

PART B — (5 × 16 = 80 marks)

11(a) (i) Explain the different types of Intelligent Agents with neat diagrams. (ii) Describe the various types of environment properties in AI.

OR

(b) (i) Discuss the structure of a Learning Agent and its components. (ii) Elaborate on the 'Turing Test' and its significance in AI history.

12(a) (i) Explain the A* Search algorithm. Prove that A* is optimal if the heuristic is admissible. (ii) Compare and contrast Uninformed Search and Informed Search strategies.

OR

(b) (i) Solve the 8-puzzle problem using Hill Climbing search. Explain the state space and the heuristic used. (ii) Describe the Uniform Cost Search (UCS) algorithm with an example.

13(a) (i) Explain the Genetic Algorithm process with its operators: Selection, Crossover, and Mutation. (ii) Discuss the 'Simulated Annealing' search technique.

OR

(b) Explain the Recursive Best-First Search (RBFS) and Memory-bounded A* (MA*) algorithms. Compare their performance in terms of space and time complexity.

14(a) (i) Illustrate the Minimax algorithm with an example game tree of depth 3. (ii) Apply Alpha-Beta pruning to the same game tree and identify the pruned branches.

OR

(b) (i) Explain how games with an element of 'Chance' (e.g., Backgammon) are handled in AI. (ii) Discuss the role of Evaluation Functions in game playing when the search tree is too deep.

15(a) (i) Formulate the Map Coloring problem as a Constraint Satisfaction Problem (CSP). (ii) Explain the Backtracking search for CSPs and the 'Minimum Remaining Values' (MRV) heuristic.

OR

(b) (i) Describe the AC-3 algorithm for maintaining Arc Consistency in CSPs. (ii) Explain the concept of

'Forward Checking' and how it helps in reducing the search space.

Faculty In-Charge

Verified by

HOD